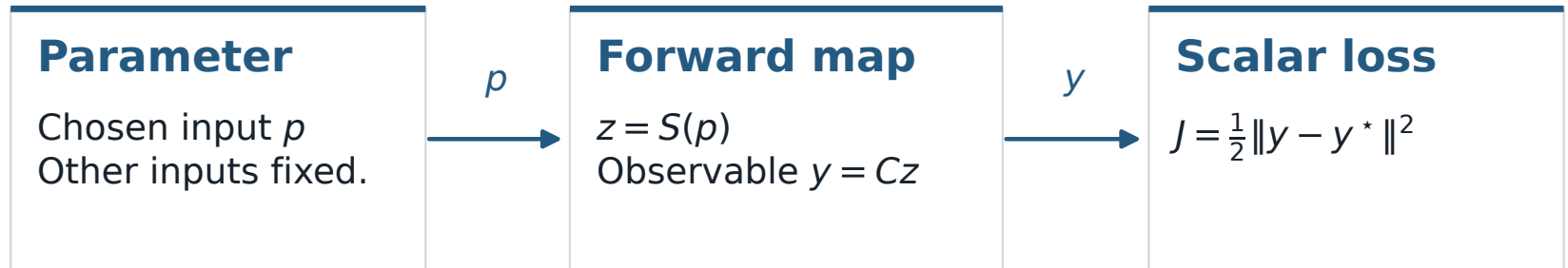
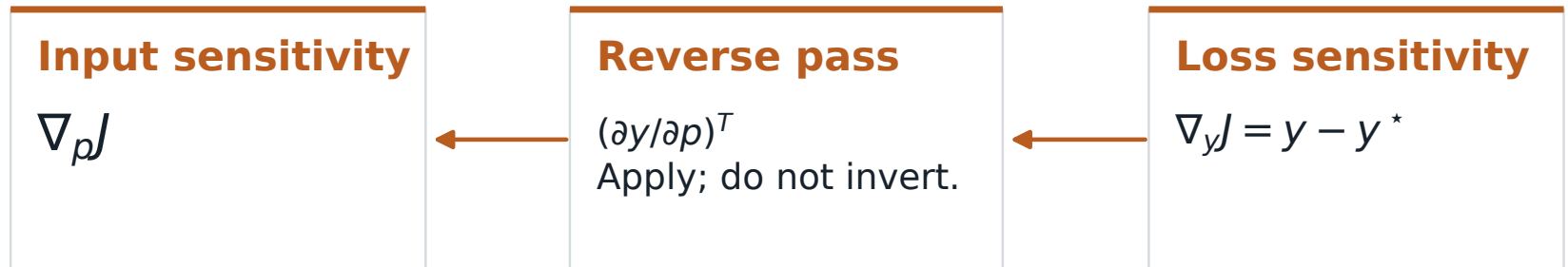


A gradient is not an optimisation step

FORWARD · evaluate the stated computation



REVERSE · accumulate vector-Jacobian products



$$\nabla_p J = (\partial y / \partial p)^T \nabla_y J$$

CHECK

$$\frac{J(p + hq) - J(p - hq)}{2h} \approx (\nabla_p J)^T q$$

OPTIONAL

$$\text{Optimise: } p_{\text{new}} = p - \alpha \nabla_p J$$

A local derivative check does not establish a unique inverse solution.